

AI item difficulty priors

AI item difficulty priors — where we are

Framing: Research tool for human researchers (priors / ranking / triage)
Goal is a *starting* difficulty for new or blank bank items before children take them, on a scale comparable to CAT bank d (higher = harder).

Shared tooling

Piece	Role
levante-qa / Cypress VLM panel (tools/vlm-panel)	Models take the real task UI → p_vlm → calibrator → p_pred_child → hybrid d_est (estimate_difficulty.mjs); optional fill of <i>blank</i> bank rows only (apply_d_est_prior.mjs)
levante-bench	Child trials / item_params for calibration + validation; also paper open-model runs as a re-runnable ability ladder (Tan et al., 2026, arXiv:2606.05497)
Lab notebook	tools/vlm-panel/lab_notebook_difficulty_estimation.ipynb

Downstream math (fit_bench_calibrator → estimate_difficulty) is the same once you have per-item success rates. **How we get those rates differs by task.**

Approach A — TROG EN: live Gemini Cypress panel

Method: Age-grid Cypress panel on current Gemini models (real TROG UI) → calibrate → hybrid d_est.

Why this works here: Panel accuracy spreads with age/persona; items differentiate.

Metric	Approx. Spearman ρ
Hybrid d_{est} vs bank d	~ 0.64
Blank items: AI vs human-linked ranking	~ 0.71
Human-IRT $\rightarrow$ bank rescale (reference)	~ 0.50

Detail: out/REPORT_d_est_initial_prior.md

Approach B — Stories / ToM: curated open-model ladder

Method: Reuse [levante-bench](#) paper open-model forced-binary runs ([Tan et al., 2026](#); Smol $\rightarrow$ Qwen / InternVL / Gemma / Molmo). Drop floor/broken zoo runs; keep a **top-5-by-ToM-accuracy** subset; ability-weighted mean success $\rightarrow$ same calibrator + `estimate_difficulty` path. Bank d is blank for ToM; validation anchors = flipped Redivis human IRT.

Why not Approach A alone: Live Gemini Cypress panel was CEILING-heavy even after force recollect (non-response 33% $\rightarrow$ 0%). Hybrid d_{est} from that panel only reached LOO $\rho \sim 0.35$ vs human IRT.

Ladder / source	Approx. Spearman ρ
All-17 open models (mean $-p$)	~ 0.26
Curated top-5, ability-weighted $-p$ vs human IRT	~ 0.70
Same ladder through hybrid d_{est} (LOO)	~ 0.61
Gemini Cypress $-p_{pred}$ / panel d_{est} (force EN)	~ 0.69 / ~ 0.35

Detail: out/REPORT_stories_paper_ladder_backtest.md,
out/REPORT_stories_d_est_paper_ladder.md

Other tasks (same pipeline, weaker signal so far)

Task	Approach	Result
Vocab EN	Mostly Approach A (panel + DIGIT YES NO / randomize-on-NO)	Hybrid ~ 0.65 ; prefer human params when present
Matrix	Approach A smoke	Weak (~ 0.20 vs bank d); may need Approach

Task	Approach	Result
		B–style ladder or better features

vs typical human *initial* difficulty guesses

(Psychometric literature—not a LEVANTE expert-guess archive yet. Domain varies; treat as a literature prior.)

- Experts are better at **ranking** items than guessing absolute % correct / IRT $\flat$ (Impara & Plake, 1998; Wauters et al., 2011; Attali et al., 2014).
- Single unaided absolute judgments often only **~0.2–0.5** with later empirical difficulty (Bejar, 1983; cf. Melican / Cross-style absolute rating studies).
- Pooled / trained / comparative methods often **~0.7–0.8+** (classic pooled ratings; training/anchors improve toward ~0.5–0.65+ in Hambleton-style work).

Our stronger AI results (TROG Approach A ~0.64; Stories Approach B ~0.70 ranking / ~0.61 d_{est}) sit in the **pooled-expert** band—and above many single seat-of-the-pants guesses—as a research prior, not final CAT truth.

Selected refs

1. Bejar, I. I. (1983). Subject matter experts' assessment of item statistics. *Applied Psychological Measurement*.
2. Impara, J. C., & Plake, B. S. (1998). Teachers' ability to estimate item difficulty. *Educational Measurement: Issues and Practice*.
3. Attali, Y., et al. (2014). Comparative judgment / ranking of item difficulty (ETS). <https://doi.org/10.1002/ets2.12042>
4. Wauters, K., Desmet, P., & Van den Noortgate, W. (2011). Expert vs data-driven item difficulty. *EDM*.
5. Also: Lorge & Kruglov (1950s); Hambleton, Bastari & Xing (training / anchors).

Repos & LEVANTE-bench paper

- [levante-qa](#)
 - [levante-bench](#)
 - Tan, A. W. M., Cardinal, D., Lorigo-Botran, T., Bravo-Sánchez, L., Yu, S., & Frank, M. C. (2026). *LEVANTE-bench: Multi-Scale Comparison of VLMs to Children Using Cognitive Tasks*. arXiv:2606.05497. <https://arxiv.org/abs/2606.05497>
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Bottom line

- **TROG:** live Gemini panel is enough.
- **Stories:** prefer the **curated open-model ladder**, not the full zoo or a ceiling-heavy Gemini panel alone.
- Useful when items differentiate; not a promote-to-GCS story.

Next: pin an open ToM ladder for new items; try the same ladder backtest on matrix/vocab; optional LEVANTE expert-guess study as a local human baseline.